$$\mathbf{M}^\dagger = \left[\mathbf{M}^T\right]^* .$$

To summarize: if **M** acts on the ket |*A*> to give |*B*>, then it follows that $\mathbf{M}^\dagger$ acts on the bra <*A*| to give <*B*|. In symbols:

If

$$\mathbf{M}|A\rangle = |B\rangle,$$

then

$$\langle A|\mathbf{M}^\dagger = \langle B|.$$

3.1.4 Hermitian Operators

Real numbers play a special role in physics. The results of any measurements are real numbers. Sometimes, we measure two quantities, put them together with an *i* (forming a complex number), and call this number the result of a measurement. But it's actually just a way of combining two real measurements. If we want to be pedantic, we might say that observable quantities are equal to their own complex conjugates. That's of course just a fancy way of saying they are real. We are going to find out very soon that quantum mechanical observables are represented by linear operators. What kind of linear operators? The kind that are the closest thing to a real operator. Observables in quantum mechanics are represented by linear operators that are equal to their own Hermitian conjugates. They are called *Hermitian operators* after the French mathematician Charles Hermite. Hermitian operators satisfy the property

$$\mathbf{M} = \mathbf{M}^\dagger.$$

In terms of matrix elements, this can be stated as